Assurance-Aware Remaining Useful Life Prediction Under Correlated Sensor Failures for Turbofan Engines

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***Abstract*—Remaining useful life (RUL) prediction must distinguish degradation from measurement failures across operating conditions. GRACE-RUL integrates local CNN–GRU features, residual BiLSTM sensor recovery, Transformer context, independent finite-fault detection and a bounded supervised corrector. One shared model pools all four NASA C-MAPSS subsets with 567 fitting, 36 selection, 35 calibration and 71 internal-test engines. Faults follow related-channel, temporally persistent mechanisms in raw sensor units. Conservative validated thresholds, fitting-only support bounds and a complete 41-condition audit limit healthy-sensor displacement. Predictor refinement uses deterministic paired-prefix continuity, fitting-error tail supervision and corruption-aware RUL supervision. A validation-selected candidate is frozen before interval calibration and test evaluation. On 707 official endpoints its capped RMSE is 12.01 cycles, MAE is 8.52, and severity-0.2 stuck-fault RMSE averages 15.08 over three corruption replicates. Internal clean RMSE is 13.30 over 484 prefixes from 71 engines. Observable interval scales are fitted independently of engine-level conformal calibration. A previously inspected severe prediction failure is reported explicitly. Under broadband sensor noise at 20% of the within-regime scale, capped RMSE rises from 12.01 to 12.37 cycles with 98.9% empirical coverage; at 50% it is 13.68 cycles. Reused benchmarks, one training seed and simulated faults limit generalization claims.**

***Index Terms*—Remaining useful life, C-MAPSS, correlated sensor faults, guarded recovery, pooled training, CNN–GRU, Transformer, conformal calibration.**

I. Introduction

Turbofan RUL estimation maps observed operating history to remaining cycles before failure. Missing sensor readings are observable, while a finite bias, drift or stuck reading can resemble degradation. Operating-condition changes make this distinction harder. C-MAPSS provides a reproducible simulation benchmark, but added measurement faults are modeling assumptions rather than recorded aircraft incidents [1].

GRACE-RUL is organized around three design commitments: local convolutional–recurrent modeling of recent degradation, selective recurrent recovery of corrupted readings, and a Transformer context branch with residual compensation. The current format-9 implementation uses one shared model across FD001–FD004, without requiring a subset label at deployment. Its contributions are a prefix-consistent related-channel fault simulator; independent finite-fault detection with selective, support-bounded repair; quality-conditioned local/global RUL fusion; validation-led predictor refinement; independently calibrated observable intervals; and a broadband measurement-noise and channel-dropout benchmark that reports accuracy and calibrated coverage without refitting. Complete recovery checks and state-bound audits make the selected path inspectable. Severe prediction failures, reused benchmarks and unvalidated fleet assumptions are retained explicitly in the evidence.

This manuscript presents the assurance-aware prognostic architecture along with empirical evaluation on reference and structured sensor-fault NASA C-MAPSS telemetry, including observed recovery limitations.

II. Related Work

A) Local Degradation Modeling and Interpretation

The supplied hybrid CNN–GRU study emphasizes local feature extraction, temporal degradation modeling, and sensor-level interpretation [2]. GRACE-RUL retains a CNN–GRU branch over the most recent 30 cycles. Interpretation is implemented as optional post-training KernelSHAP/LIME analysis [5], [6]; it is not a term in the neural training objective. That work's complete reliability/generalization/explainability formulation is not reproduced literally.

B) Recovery Under Partial Sensor Failure

The partial-failure study motivates paired corrupted/reference histories, bidirectional recurrent recovery, and a training-only adversarial critic [3]. The implementation adds a deployment-available finite-fault detector and availability-based selective blending, because synthetic altered-entry labels are unavailable at inference. WGAN-GP supplies the adversarial training objective [7]. Original C-MAPSS readings remain noisy reference measurements, rather than pristine sensor truth.

C) Transformer Context and Residual Compensation

Trans-D3 motivates combining a Transformer representation with prediction correction [4]. The current implementation instead uses an OOF-trained supervised residual MLP with

bounded output together with physical degradation slew and anti-rebound loss penalties on paired engine prefixes. It contains no TD3 actor, twin action-value critics, replay buffer, policy update, or LQR filter. The separate WGAN critic evaluates sensor sequences during recovery training and is unrelated to reinforcement learning.

III. Problem Statement and Data Protocol

A) Observed Histories, Targets and Masks

An observation contains three operating settings and 21 sensors. Engine ID and cycle order trajectories; neither is a neural feature. A window ending at t contains at most $L = 70$ already observed cycles, left-padded to fixed length. Validity V masks padding and availability M marks finite sensor readings. Normalized settings are U and encoded observed sensors are S; nonfinite measurements are replaced by zero after the mask is constructed. A zero measurement remains valid. For a complete fitting engine ending at T_e, raw RUL is $r_{e,t} = T_e - t$ and

$$y_{e,t} = \min(125, r_{e,t}). \tag{1}$$

Official RUL labels enter evaluation after freeze. Targets, future observations, source identity and simulated altered-entry labels never enter inference. Softplus estimates may exceed 125: the cap specifies the target, not a hard prediction ceiling or a guarantee about future failure time.

B) One Pooled Engine-Disjoint Split

The available four training files are pooled after namespacing engine IDs for bookkeeping. Split seed 42 assigns approximately 80% to fitting, 10% to a selection/calibration pool and 10% to internal test. Half of the development pool is reserved for independent interval calibration. Exact rounded counts are 567 fitting, 36 selection, 35 calibration and 71 internal-test engines. Every prefix of an engine stays in its partition. Per-FD reporting is an evaluation breakdown, not a collection of different models.

C) Fitting-Only Regimes and Encoding

Operating settings are standardized using fitting engines only. GMM BIC selects the regime count from fitting settings, followed by K-means assignment; the selected shared scaler has 6 regimes. Per-regime sensor means/scales include measurement-resolution floors. For raw sensor x_j in assigned regime k,

$$z_j = (x_j - \mu_{kj})/\sigma_{kj}, \tag{2}$$

$$e_j = s\,\mathrm{asinh}(z_j/s), \qquad s = 1, \tag{3}$$

$$\phi^{-1}(e_j) = s\sinh(e_j/s). \tag{4}$$

This invertible encoding compresses extreme standardized values; it does not reduce their physical raw-unit error. Faults are applied after decoding and only changed cells are re-encoded. Intact encoded cells retain their bits. Recovery support is the empirical per-regime/channel fitting range, not a certified manufacturer operating envelope. Selection, calibration and test engines fit none of these statistics.

D) Causal Prefix Evaluation

Fitting uses engine-balanced sampling from eligible within-engine windows. Selection and internal testing use a fixed inspection grid, beginning at cycle 30 with stride 30, independent of eventual failure. Calibration evaluates every observed prefix of each reserved complete engine. Official test evaluation uses its final observed endpoint. Bidirectional recovery and non-triangular attention operate inside the already observed window; unseen future readings are absent. The local branch uses 30 cycles and the global branch 70. Neither enforces monotonic predictions at inference.

E) Data Integrity and Evidence Boundaries

The shared model receives only normalized operating settings, observed sensors, availability and the observed-history mask. Engine IDs, subset labels, future readings, true RUL and simulated altered-entry masks are excluded from inference. Related-channel fault histories are prefix-consistent: appending future rows does not alter an earlier corruption realization or select onset by eventual failure.

The four original training files define the engine-disjoint fit/selection/ calibration/internal-test groups in Table I. Scaler, raw-unit fault scales and repair support use fitting engines only. Fitting-engine OOF features exclude their held fold from temporary fitting and inner selection. Predictor refinement uses fitting engines; every clean/source/NASA, mixed/stuck and tail-selection check uses held selection engines. Calibration engines enter only after that choice is fixed.

New artifacts use fresh directories and retain their source SHA-256. Resuming an interrupted refinement requires matching source, recipe and implementation. Current detector audits bind the actual deployed thresholds and detector state; the earlier detector operating point remains historical metadata. Recovery proofs bind the full declared context matrix, selection IDs and sensor-repair state. Calibration binds the final model/configuration and its separate scale scope. Strict reload rejects stale or mismatched records.

Validation-only development runs block official test/RUL and saved test-result reads. These checks document within-run separation. Earlier internal and official test cases have already informed development; their later evaluation remains exploratory and is not an untouched independent generalization test.

IV. Structured Sensor and Acquisition Faults

Added corruption models measurement failure while retaining original degradation and RUL labels. A profile selects a related measurement domain: temperature, pressure, fan speed, core speed, coolant flow, or an individual derived quantity. These groupings are measurement-based assumptions, not verified wiring diagrams. Operating settings and the two speed-demand columns are preserved.

Mixed mechanisms share a domain, channels, and onset. Regime transitions causally increase noise amplitude and burst-entry risk through a stress proxy computed from already observed settings. Bias/drift bounds scale with $2.5v$ times the

TABLE I
CURRENT POOLED ENGINE-DISJOINT DATA PARTITIONS.

Subset	Available training	Fit	Selection	Calibration	Internal test	Official test
FD001	100	80	5	5	10	100
FD002	260	208	13	13	26	259
FD003	100	80	5	5	10	100
FD004	249	199	13	12	25	248

The 10% development pool is divided equally between selection and independent calibration; integer rounding gives the exact counts above. Internal observations are fixed-grid prefixes, not independent engines.

TABLE II
IMPLEMENTED FAULT MECHANISMS. SEVERITY IS NOT A MISSING-CELL PERCENTAGE.

Family	Temporal and channel structure
Intermittent	Related-channel burst dropout; minimum two cycles, configured mean four.
Outage	Persistent channel unavailability after onset.
Block	Recurring missing blocks with intact gaps of 25–50 cycles by default.
Noise	AR(1) error, $\rho = 0.9$; common-source and individual innovation variance weights 0.8/0.2.
Bias	Persistent additive calibration offset in raw sensor units.
Drift	Raw-unit offset ramps over 100 cycles to its bounded level.
Stuck	A physical reading freezes at onset, including across regime transitions.
Mixed	Connector: noise/intermittent; calibration: bias/drift; acquisition: stuck/block.

fitting fault scale; noise amplitude also depends on v. Onset is an observed offset between zero and 100 cycles, independent of eventual failure time. Seeded streams preserve prefixes when future rows are appended, and cached histories give overlapping windows identical corruption.

Randomness represents uncertain occurrence within a specified fault process. It is temporally and physically structured rather than independent arbitrary perturbation of every entry. Training samples $v \in \{0.1, 0.2, 0.3\}$ with a ten-epoch curriculum. The default clean-realization probability is 0.25, not a batch quota. The historical format-5 study used selected subset overrides. Severities 0.4 and 0.5 extend beyond the default sampled training levels. A stuck reading can generate a large error after a regime transition even at $v = 0.2$. No severity parameter has been calibrated to field failure rates.

V. CURRENT FRAMEWORK ARCHITECTURE

Fig. 1 separates inference from training-only supervision. A single pooled parameter set serves all four source subsets; operating settings condition the computation. The recovery guard adds fitting-only buffers and a deterministic projection.

A) Independent Finite-Fault Detector

The detector receives $[S, M, U]$ (45 features per cycle). Parallel kernel-three and kernel-five convolutions each produce 32 ReLU channels. A GRU-64 and linear 21-logit head yield sigmoid discrimination scores p. The head starts with zero weights and bias -4, so its initial score is approximately 0.018. A constant initialized head is not reported as learned fault evidence.

Only finite, real entries train the classifier. For each sensor, binary cross entropy is averaged separately over present altered and intact classes, those present-class means are averaged, and sensor losses are then averaged. Missing readings do not become finite-fault positives. The detector is fitted independently on fitting engines using clean and all fault/mixed contexts. Validation selects its checkpoint and representable sensor thresholds τ_j, subject to at most 5% empirical intact false positives in each validation channel/context. Scores are not calibrated physical failure probabilities.

The deployed gate retains a conservative threshold floor and tightens only channels that violate a source/context false-positive bound on held selection engines. Its current audit binds the actual thresholds and detector state; previous detector metrics remain explicitly historical. The detector is frozen during generator, joint and predictor-refinement training. Detached scores drive the generator and recovery gates, so RUL/reconstruction gradients cannot redefine fault classification. Threshold $\tau_j = 1$ explicitly disables finite repair, including when sigmoid rounds to one.

B) Residual Generation and Thresholded Proposal

A two-layer bidirectional LSTM with hidden size 64 per direction receives $[S, M, p, U]$ (66 features). A $128 \to 64 \to 21$ ReLU decoder, initialized with a zero final layer, produces

$$G = S + f_{\mathrm{BiLSTM}}([S, M, \mathrm{sg}(p), U]), \tag{5}$$

$$\bar{p}_j = \mathrm{sg}(p_j)\, \mathbf{1}\{p_j \geq \tau_j, \tau_j < 1\}, \tag{6}$$

$$W = (1 - M) + M \odot \bar{p}, \tag{7}$$

$$R^{\star} = (1 - W) \odot S + W \odot G. \tag{8}$$

Here sg stops gradients. Missing entries route to G; finite readings require the selected detector gate. The generator residual is unconstrained internally, but the deployed finite proposal is subsequently bounded. A bypass sets $R = S$ and supplies zero recovery weights to the predictor. All stages exclude padding.

C) Fitting-Support Guard for Finite Readings

Let $x = \phi^{-1}(S_j)$ be a finite observed standardized reading, and let $[\ell, u]$ be its fitting support in the selected regime. With

existing displacement budget $\varepsilon = 0.15$, define

$$\mathcal{I}(x;\ell,u) = \begin{cases} [x-\varepsilon, x+\varepsilon], & \ell \le x \le u, \\ [\ell-\varepsilon, x+\varepsilon], & x > u, \\ [x-\varepsilon, u+\varepsilon], & x < \ell. \end{cases} \tag{9}$$

The finite result is

$$R_j = \phi\big(\mathrm{clip}(\phi^{-1}(R_j^\star), \mathcal{I}(x;\ell,u))\big)\,. \tag{10}$$

Bounds round inward to representable values, preserving identity proposals. An observed inlier therefore moves at most ε in original-linear standardized coordinates. An outlier may return throughout the fitting range in one step; outward movement remains budgeted. Missing imputation is unchanged. Projection uses ordinary clamp gradients without a straight-through estimator.

This is a conditional guarantee about an observed inlier, including a genuinely corrupted reading that happens to lie inside the support. It can restrict useful small-fault correction. A healthy unseen reading can lie outside fitting support and receive a larger inward move. Thus the rule is neither a universal healthy-cell guarantee nor a raw-unit tolerance of 0.15. Raw displacement depends on σ_{kj}.

D) Local and Global Prognostic Branches

The predictor receives $[R, U, M, W]$ (66 features). Its local branch uses the last 30 cycles: parallel kernel-three/five convolutions with 32 channels each, GRU-64, and a $64 \to 64$ last-valid-output projection. Repaired local values may depend on the entire observed recovery window. The global branch projects the 70-cycle input to 64 dimensions, adds sinusoidal positions, and uses two Transformer encoder layers with four heads, feed-forward width 128, GELU and dropout 0.2. Padding is masked; the last valid token represents global context.

E) Quality Fusion and RUL Heads

The quality vector $q \in \mathbb{R}^{64}$ concatenates time-mean availability (21), recovery weights (21), absolute actual changes $|R - S|$ (21), and observed-history fraction (one). A $192 \to 32 \to 2$ gate combines local and global features:

$$(g_\ell, g_g) = \mathrm{softmax}(f([h_\ell, h_g, q])), \tag{11}$$

$$h = g_\ell h_\ell + g_g h_g, \tag{12}$$

$$\hat{y}_0 = 125\,\mathrm{softplus}(f_{128\to64\to1}([h, q])). \tag{13}$$

The main head has ReLU/dropout 0.2. Separate 64-to-one auxiliary heads supervise the two branches. Fusion coefficients are learned mixing weights, not certified reliability estimates.

F) Observable OOF Bounded Corrector

The corrector receives q, normalized base RUL, and the two normalized auxiliary predictions: 67 observable features. A $67 \to 32 \to 1$ ReLU MLP gives

$$\Delta = b\tanh(f_c(\cdot)), \qquad \hat{y} = \max(0, \hat{y}_0 + \Delta). \tag{14}$$

Validation selects $b \in \{5, 10, 15, 30\}$ or a zero-correction fallback. This bounds the adjustment, not total RUL or temporal prediction changes. The corrector receives neither labels nor engine indices as features. OOF construction excludes held engines from temporary upstream fitting and inner validation; independently learned hidden embeddings are excluded because their coordinate bases need not agree.

G) Training-Only Critic and Model Size

A conditional critic consumes reference/repaired sensors plus settings (24 features). Kernel-three convolutions $24 \to 32 \to 64$ with LeakyReLU 0.2 precede LSTM-64 and a scalar head. It supplies WGAN-GP feedback during training and is absent at deployment. Detector, generator, predictor and corrector contain 37909, 176533, 132005 and 2209 learned parameters, respectively: 348656 shared parameters in total. The separate critic is absent at inference. Support and thresholds are fitted buffers rather than additional learned parameters.

VI. Training and Validation-Led Repair

A) Source Training and Selective Recovery

Clean pretraining bypasses recovery and correction. A separate detector learns finite-fault discrimination from fitting-engine labels; missing entries are not finite-fault positives. Detector weights are then frozen. Paired recovery uses changed-entry reconstruction, healthy identity and temporal terms, plus a conditional WGAN-GP critic. Its discriminator objective is

$$\mathcal{L}_D = \mathbb{E}D(R,U) - \mathbb{E}D(S^{\mathrm{ref}},U) + 10\mathcal{L}_{\mathrm{GP}}. \tag{15}$$

The generator adversarial objective is $-\mathbb{E}D(R,U)$. These source stages also supervise RUL and auxiliary heads. A fitting-only support projection bounds finite proposals; sensor labels supervise training, not the deployed gate.

Three fitting-engine OOF folds supply observable correction features. Each held fold is excluded from its temporary scaler/upstream fitting and inner selection. The bounded MLP corrector is a supervised adaptation of residual compensation; it is not a TD3 policy. The source corrector is frozen and revalidated during predictor refinement. Its historical OOF training does not establish nested cross-validation of the changed final pipeline.

B) Deterministic Continuity and Fitting-Error Tails

Predictor refinement freezes detector, generator and corrector parameters. It uses clean reference RUL supervision and the exact active forward path on structured corrupted fitting windows. For two observed prefixes $t_1 < t_2$ of one fitting engine, both sides of the continuity objective use deterministic passes, avoiding independent dropout as a source of artificial trajectory change. Labels anchor both predictions

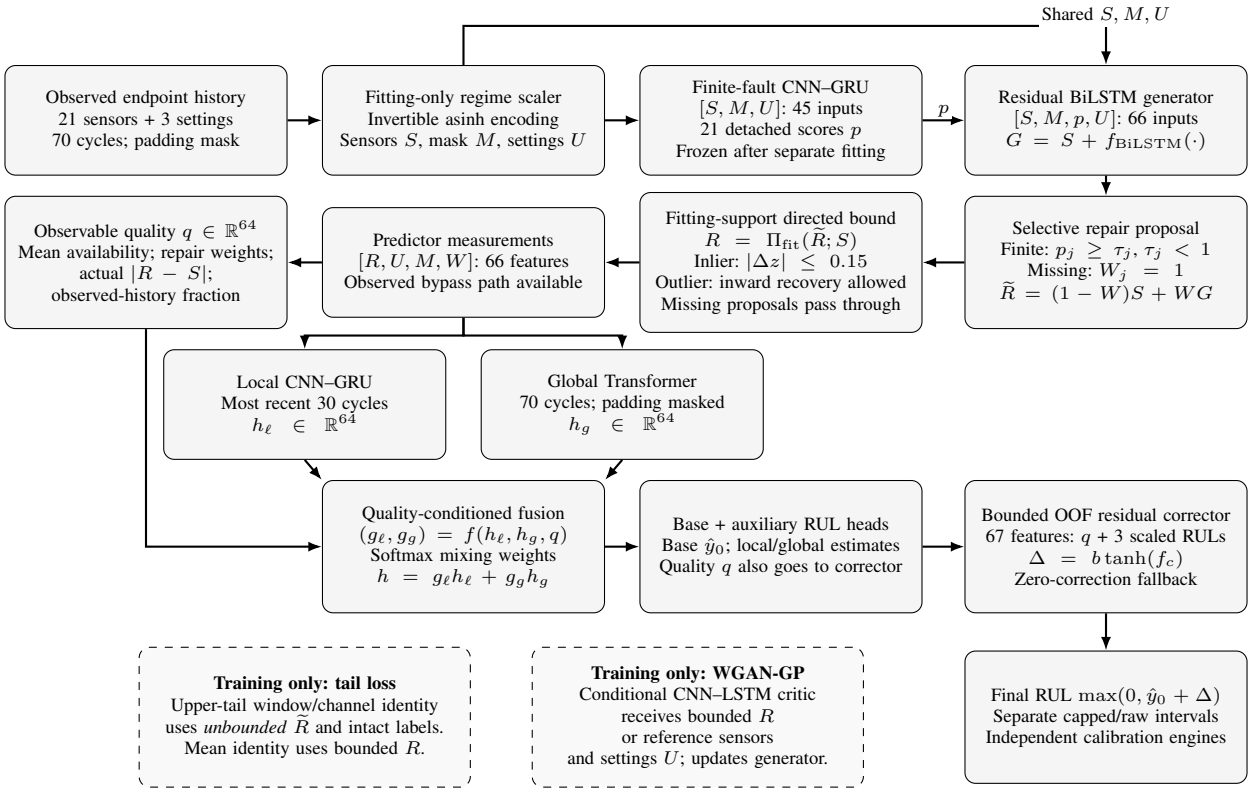


Fig. 1. Shared format-9 computation for every source subset; a subset label is not an inference input. The directed bound acts in original, uncompressed regime-standardized sensor coordinates z, using fitting-only support. The 0.15 budget applies to observations inside that support; large outliers can move inward without a universal small-step cap. The detector and generator remain frozen during the reported predictor refinement. Dashed components describe source recovery training, and the WGAN critic is absent at deployment. Deterministic paired-prefix and fitting-tail objectives refine the predictor without adding a hard inference floor or monotonic constraint. Altered-entry labels and targets never enter deployment.

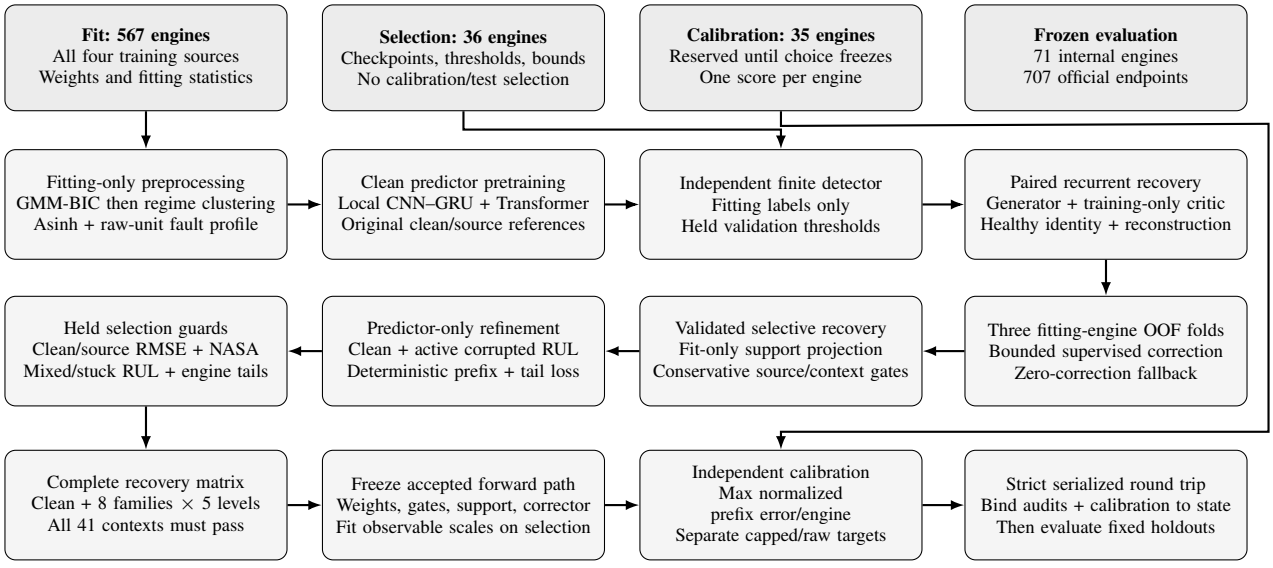


Fig. 2. Current pooled workflow. Approximately 80% of available training engines fit weights; the 10% development pool is divided between selection and calibration, and 10% is internal test. Exact rounded counts are shown. Detector/generator/corrector weights are frozen during predictor refinement; source OOF correction is retained and revalidated. Rejected candidates produce no deployable export. Validation-only pilots precede any new calibration or holdout evaluation.

and their capped-target difference. Additional soft penalties target excessive downward slew and upward rebound:

$$\mathcal{L}_{\mathrm{slew}} = \left[\frac{(\widehat{y}_1 - \widehat{y}_2) - (t_2 - t_1)}{125} \right]_+^2, \tag{16}$$

$$\mathcal{L}_{\mathrm{rebound}} = \left[\frac{\widehat{y}_2 - \widehat{y}_1}{125} \right]_+^2. \tag{17}$$

These are regularizers, not validated physical laws or hard inference constraints. They permit later predictions to recover from an erroneous early estimate.

A fitting-only tail objective averages the largest fraction of normalized squared batch errors, rather than assigning a special engine an output floor. The refinement combines clean RUL, auxiliary-branch, paired-prefix, tail and active corrupted-RUL losses. Its weights and draw budget are serialized with optimizer/scheduler/RNG state in resumable epoch checkpoints. No true target, engine-specific rule or hand-assigned accuracy enters deployment.

C) Thresholds and Candidate Acceptance

A declared conservative finite threshold floor is preserved. For a violating source/channel/context, the smallest representable score boundary that satisfies the empirical false-positive guard tightens that channel; already disabled channels stay disabled. This preserves useful repair opportunities while exposing source-specific false alarms. Detector scores are discrimination scores, not calibrated probabilities of physical failure.

Every candidate must keep pooled and per-source clean RMSE/NASA within the original 3%/10% budgets and retain sensor-reconstruction/intact guards. Mixed and stuck RUL also remain within 3% of the preserved source, pooled and per source. Selection minimizes the mean worst-decile engine maximum-prefix absolute error, with clean RMSE as a tie-breaker. Source references remain fixed; each stage cannot spend a new tolerance. The complete recovery matrix contains clean once plus eight fault/mixed families at severities 0.1–0.5, totaling 41 contexts. Incomplete or rejected scans cannot certify an export. All diagnostics and failure flags are retained.

The selected predictor-refinement epoch is 14; the validation engine-tail score changes from 33.221 to 33.064 cycles. Every one of the 24 final active-path guards passes. The deployed detector audit uses the actual saved thresholds, with clean selection FPR 0.0322%. Its earlier training operating point is retained as historical metadata. The 41-context recovery audit has worst pooled-context intact original-linear RMSE 0.00983 and worst window RMSE 0.04419. The reported refinement uses AdamW with learning rate 0.00002, weight decay 0.0001, batch size 128, and a cosine schedule to 0.00001. The budget is 16 epochs and 8192 fitting draws per epoch. Clean, auxiliary-branch, prefix, tail and active corrupted-RUL weights are 1, 0.1, 1, 1 and 2; the tail fraction is 0.1. Slew and rebound weights inside the prefix term are 1 and 1.

D) Freeze, Calibration and Strict Export

The repaired policy uses observable scaling with one score per independent engine, following the split-conformal construction [8]. For each target convention $h \in \{c, r\}$, first fit a positive scale $a_h(\widehat{y})$ on selection engines, independently of calibration. Observable predicted-RUL bins are $[0, 35)$, $[35, 75)$, $[75, 125)$ and $[125, \infty)$. Within a supported bin, the scale is selection RMSE, with a positive one-cycle floor. An unsupported scale bin uses global selection RMSE. This is a fixed normalizer, not a replacement for an unsupported calibration quantile. True RUL is used as a selection/calibration label, never to choose an inference bin.

After freezing weights, support, thresholds, corrector and scales, evaluate all observed prefixes of each of the 35 reserved engines. For capped $y^{(c)} = \min(r, 125)$ and raw $y^{(r)} = r$, form

$$s_e^{(h)} = \max_{t \in \mathcal{P}_e} \frac{|y_{e,t}^{(h)} - \widehat{y}_{e,t}|}{a_h(\widehat{y}_{e,t})}. \tag{18}$$

The maximum spans every bin visited by an engine, so dependent prefixes and observable bins do not become extra independent calibration samples. With n engines, $\alpha = 0.1$ and $k = \lceil (n+1)(1-\alpha) \rceil$, define

$$q_h = \begin{cases} s_{(k)}^{(h)}, & k \le n, \\ +\infty, & k > n. \end{cases} \tag{19}$$

At inference, the observable adaptive interval is

$$C_h(\widehat{y}) = \big[\max\{0, \widehat{y} - q_h a_h(\widehat{y})\},\ \widehat{y} + q_h a_h(\widehat{y}) \big]. \tag{20}$$

An unbounded rank remains unbounded; it is not silently replaced by a finite global radius. The saved quantile, independent scale-engine scope, bin radii and model/configuration fingerprint are checked during loading.

For each target convention separately, exchangeable complete engine trajectories and a model/normalizer fixed independently of calibration give marginal simultaneous-prefix coverage across all observable bins. This is an adaptation of split conformal to engine-level scores, not a claim of conditional coverage for an individual engine or a life stage. It gives no joint guarantee across capped and raw contracts. Censoring, fleet shift and added faults need not preserve the assumptions. Empirical official endpoint coverage and internal prefix/whole-engine coverage are reported separately. The final artifact stores matching model/bundle configurations, thresholds that match gate buffers, current detector metrics, fitting-only support and a complete state-bound recovery audit. Calibration is bound to the full model/configuration. Strict serialized loading precedes test evaluation. New runs use fresh output directories; a source bundle is never overwritten to manufacture a comparison.

VII. Evaluation and Reproducibility

The current study uses one shared seed-42 pipeline. Official endpoints and internal fixed-grid prefixes are distinct partitions. For $d_i = \widehat{y}_i - y_i$,

$$\mathrm{RMSE} = \sqrt{N^{-1} \textstyle\sum_i d_i^2}, \tag{21}$$

$$\mathrm{MAE} = N^{-1} \textstyle\sum_i |d_i|, \tag{22}$$

$$S = \textstyle\sum_i \begin{cases} \exp(-d_i/13) - 1, & d_i < 0, \\ \exp(d_i/10) - 1, & d_i \ge 0. \end{cases} \tag{23}$$

Capped and raw targets have separate point/interval contracts; a capped interval does not imply raw-RUL coverage. The asymmetric NASA fleet total and mean are kept distinct. Fault

families are evaluated at five severities and three seeded corruption realizations with the same frozen weights. Corruption replicates are not independent training seeds or fleet samples. Engine-balanced and prefix metrics are identified in saved records. Software tests establish implementation behavior, not field performance or superiority over the source papers.

VIII. Results and Failure Analysis

A) Clean Accuracy and Evaluation Units

The frozen shared model achieves official clean capped RMSE 12.0080 and MAE 8.5224 cycles across 707 endpoint engines. Original pooled revision 2 had RMSE 12.1062 on the reused official benchmark. Internal clean RMSE is 13.2974; its 484 prefixes belong to 71 engines and are dependent within engine. These comparisons describe completed runs, rather than an independent confirmatory test. The immediate preserved fine-tuned source had official RMSE 11.9415; Table IV reports that comparison as well, including any regressions. Validation guards and the held selection-tail objective fixed the current model before these test values were read.

B) Structured-Fault Robustness

At severity 0.2, mean pooled official stuck-fault RMSE is 15.0767 cycles over three corruption realizations. Averaging all declared stuck severities and realizations instead gives 17.6636; these summaries must not be interchanged. Fig. 4 exposes severity dependence on the pooled fleet and the multi-regime subsets. Corruptions preserve original RUL labels and do not represent recorded aircraft sensor incidents.

C) Calibration and Healthy-Sensor Impact

Empirical clean coverage is 99.29% on official endpoints and 98.97% on internal prefixes. Internal whole-engine coverage over the evaluated grid is 95.77%, a different unit from prefix coverage and from every-cycle trajectory coverage. Nominal coverage is 90%, conditional on the stated exchangeability assumptions. The complete recovery matrix is accepted with worst pooled-context intact original-linear RMSE 0.00983; this is a sensor displacement diagnostic, not a RUL accuracy guarantee.

D) Broadband Sensor Noise and Channel Dropout

The structured protocol models failures confined to related channels. Real acquisition also carries broadband noise on every channel at once and occasional missing cells. We therefore add an independent measurement layer that perturbs raw sensor readings before the frozen fitting-only scaler: per-channel Gaussian noise scaled to the within-regime sensor scale, plus independent cell dropouts. Settings and speed commands stay untouched, and the layer is applied only after freeze, so no weight, detector threshold, corrector or interval radius was fitted on it. Levels are nested: one per-engine draw is scaled, so a sweep is paired rather than resampled.

Table VII shows graceful degradation. At 20% of the within-regime scale, pooled official RMSE moves from 12.01 to 12.37 cycles, an increase of 3.0%, and empirical coverage remains 98.9%. At 50% it is 13.68 cycles with 99.2% coverage. The internal partition follows the same pattern, from 13.30 to 15.44. Modest dropout is not harmful: dropping 2% of cells leaves pooled official RMSE at 11.96, and 5% gives 12.23. These are sensitivity measurements of one frozen artifact, not a claim that any level matches a measured sensor specification.

E) Comparison with the Supplied Studies

Table VIII transcribes the reported values of the supplied studies. This work is first on FD002 among every reported row, second on FD004 behind the reported TRANS-D3 value of 12.26, third on FD003 among third-party baselines, and 6th on FD001. It is not first on FD004 and no such claim is made. The hybrid CNN–GRU study reports 13.1, 16.8, 14.5 and 18.9, so this model improves every subset by 1.17 to 6.33 cycles at a comparable 125-cycle target.

Two qualifications matter more than the ordering. First, the compared entries are subset-specific models, whereas this work uses one shared model without a subset label, so the comparison is not budget-matched in the harder direction. Second, the TRANS-D3 FD001 and FD003 values of 1.66 and 6.61 lie far below every third-party entry in the same table, whose best values are 10.61 and 10.71; no capped C-MAPSS protocol in this study reaches them, so they are retained for completeness rather than treated as a reproducible target. The partial-failure study reports a single pooled fault-conditioned RMSE of 12.15 with unspecified channel coverage and severity, which is not decomposable by subset and is therefore not entered as a per-subset cell.

F) Component Contribution and Interval Width

Table IX separates what the frozen artifact actually gains from its parts. The guarded recovery path is the only component with a material effect, and only under persistent faults: stuck-fault RMSE falls from 20.07 to 15.08 cycles (4.99) at severity 0.2, the gain stays between 3.62 and 4.99 across severities 0.1–0.5, and mixed faults improve by 1.38. Outage and block faults gain about 0.3 cycles. Broadband noise gains nothing measurable, which is expected because recovery targets finite abnormal readings rather than additive measurement error.

The bounded observable corrector does not earn its place on accuracy: removing it changes RMSE by at most 0.03 cycles in any tested condition, and never for the worse. On clean endpoints the full path gives 12.01 against 11.98 for the branch-fusion predictor with neither intervention. The corrector is retained because it belongs to the frozen artifact whose validation guards passed, not because it lowers RUL error, and the manuscript does not claim otherwise.

The same table pairs coverage with sharpness. The mean clean official interval is 68.2 cycles wide against a 125-cycle target, so 99.3% empirical coverage at this radius is driven by width rather than by accurate point prediction. Coverage must therefore be read alongside width, and neither establishes conditional coverage for a realized engine. Component selec-

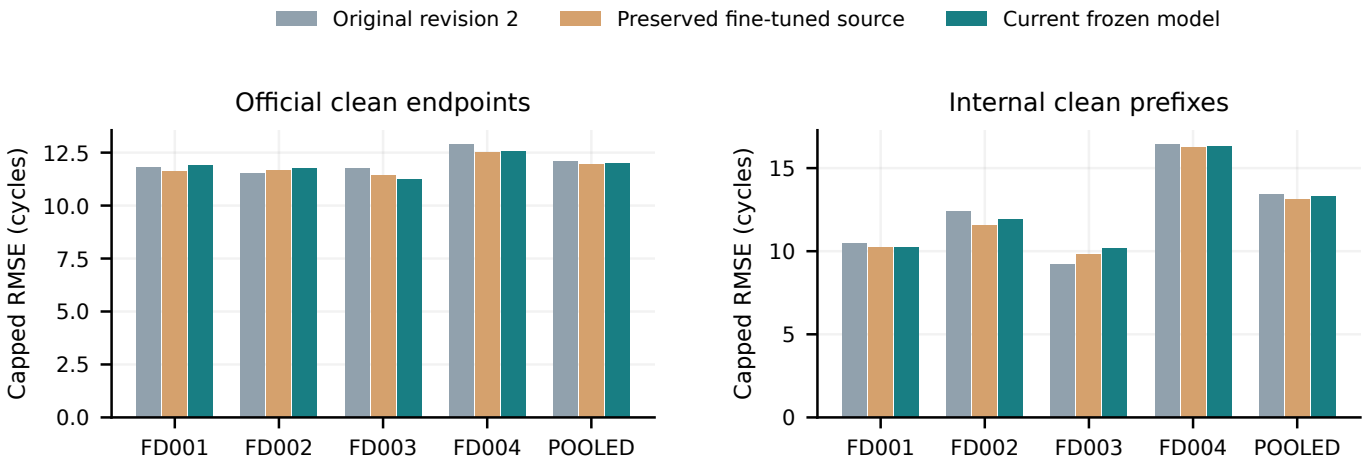


Fig. 3. Clean capped RMSE for the frozen current model, its immediate preserved source, and original pooled revision 2. Each is one training-seed pipeline; no across-training-seed uncertainty is implied. Internal values use a fixed causal prefix grid.

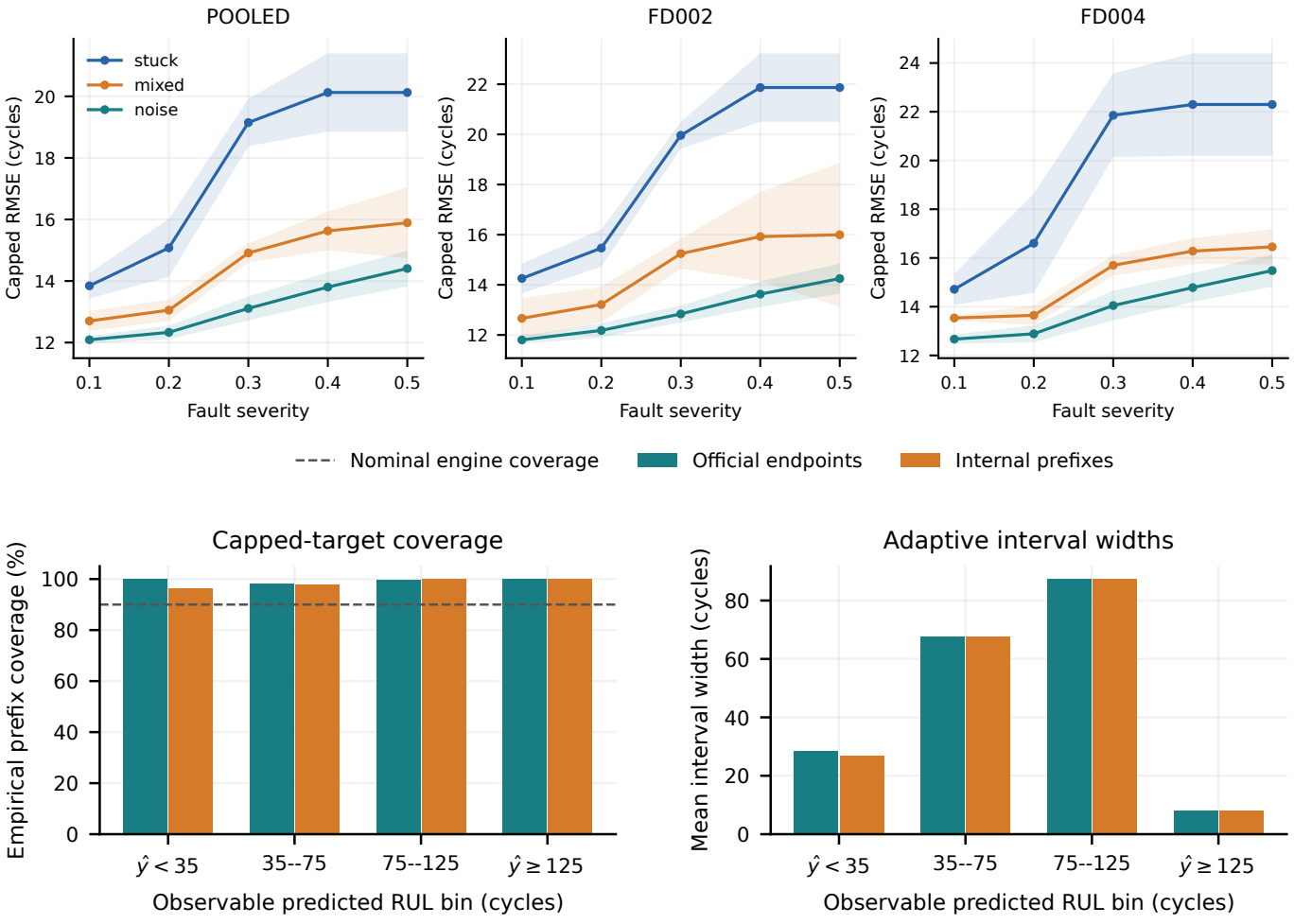


Fig. 4. Top: official capped-target robustness under structured faults; curves are means and bands are sample standard deviations over three corruption replicates of one frozen model, not training seeds or fleet confidence intervals. Bottom: empirical clean coverage and mean interval width grouped by observable predicted RUL. Grouped coverage is descriptive and does not establish conditional coverage for a realized engine or life stage. The simultaneous calibration score spans all bins for each engine.

TABLE III
CURRENT FORMAT-9 MODEL: CAPPED-TARGET CLEAN AND STRUCTURED-FAULT RESULTS.

Subset	N_{off}	RMSE	MAE	NASA S	Cov. (%)	Stuck	Mixed	Internal RMSE
POOLED	707	12.01	8.52	1671.0	99.3	15.08	13.05	13.30
FD001	100	11.93	8.85	194.0	100.0	12.27	12.36	10.23
FD002	259	11.77	8.47	541.1	99.6	15.47	13.21	11.91
FD003	100	11.25	7.92	234.3	100.0	12.24	11.70	10.18
FD004	248	12.57	8.69	701.6	98.4	16.60	13.65	16.31

Fault columns average three corruption replicates at severity 0.2; these are not training-seed replications. Coverage is empirical clean prefix/endpoint coverage. Internal RMSE uses all saved prefixes from each source's reserved engines. Targets are $\min(r, 125)$; units are cycles except NASA score and coverage.

TABLE IV
POOLED COMPARISON INCLUDING THE IMMEDIATE PRESERVED SOURCE.

Artifact	Official clean RMSE	MAE	Internal clean RMSE	Stuck @0.2	Stuck all levels
Original revision 2	12.106	8.550	13.441	18.858	20.571
Preserved fine-tuned source	11.942	8.228	13.128	15.206	17.879
Current frozen repair	12.008	8.522	13.297	15.077	17.664

All rows use their own saved predictions and capped targets. Fault values are arithmetic means across declared corruption realizations; the last column also averages severities. Reused test results are descriptive and do not select the current candidate.

TABLE V
CLEAN OFFICIAL RESULTS AGAINST ORIGINAL, UNCAPPED RUL.

Subset	Raw RMSE	Raw MAE	Raw NASA S	Raw coverage (%)
POOLED	22.34	14.69	9201.9	99.3
FD001	13.34	9.92	226.9	99.0
FD002	24.59	15.96	4660.4	99.6
FD003	13.11	9.48	277.2	100.0
FD004	25.49	17.38	4037.4	98.8

The point model was trained on capped targets. Raw errors and separately calibrated raw intervals expose long-horizon limitations; capped coverage must not be presented as raw-target coverage.

TABLE VI
OBSERVABLE INTERVAL SCALES AND RADII OF THE FROZEN MODEL.

Predicted RUL bin	Scale capped	Scale raw	Radius capped	Radius raw	Scale engines
$[0, 35)$	3.86	3.86	15.14	14.23	36
$[35, 75)$	8.60	8.60	33.69	31.66	36
$[75, 125)$	11.14	81.97	43.68	301.93	36
$[125, \infty)$	1.00	175.52	3.92	646.53	24

Scales use selection engines only. A single engine-maximum normalized score across all bins is calibrated on 35 separate engines; alpha=0.1, rank=33. Scale-engine counts are overlapping bin support, not independent calibration sample counts. No true RUL is required to choose a bin at inference.

TABLE VII
FROZEN-ARTIFACT ACCURACY UNDER BROADBAND SENSOR NOISE AND CHANNEL DROPOUT.

Noise	Official RMSE	Official MAE	Cov. (%)	Internal RMSE	Cov. (%)
none	12.01	8.52	99.3	13.30	99.0
5%	12.05	8.60	99.3	13.37	99.0
10%	12.17	8.62	99.2	13.55	99.2
20%	12.37	8.76	98.9	13.84	98.8
30%	12.67	9.04	99.0	14.29	98.8
50%	13.68	9.97	99.2	15.44	97.9
2% cells	11.96	8.56	99.6	13.37	99.0
5% cells	12.23	8.77	99.2	13.54	99.0

Rows 1–6 add per-channel Gaussian noise; the level is a multiple of the fitting-only within-regime sensor scale. Rows 7–8 instead drop whole cells at zero added noise. Settings and speed commands are unperturbed. Noise is applied after model freeze; no weight, threshold, corrector or interval radius was fitted on these levels. Coverage is empirical clean endpoint/prefix coverage under the frozen radius, not a new guarantee.

tion was frozen before these values were read; the table is a post-freeze diagnostic and the deployed artifact is unchanged.

G) Recovery and Detection Benchmark

Table X opens the recovery black box. Detection is deliberately conservative: precision exceeds 0.86 in every finite fault

TABLE VIII
REPORTED C-MAPSS RESULTS AT THE 125-CYCLE TARGET AGAINST THE SUPPLIED STUDIES.

Model	FD001	FD002	FD003	FD004
GRACE-RUL (one shared model, no subset label)	**11.93**	**11.77**	**11.25**	**12.57**
Hybrid CNN–GRU (reported, five repeats)	13.1	16.8	14.5	18.9
TRANS-D3 (reported)	1.66	13.23	6.61	12.26
Best third-party baseline (reported)	10.61 (STAR)	13.47 (STAR)	10.71 (STAR)	15.86 (DLformer)
Reported value ahead of GRACE-RUL	TRANS-D3 1.66	none	TRANS-D3 6.61	TRANS-D3 12.26
Position among third-party baselines (proposed row excluded)	6	1	3	1

Reported columns are transcribed from the cited papers and are not recomputed here. Rank counts only third-party baselines, so a proposed row cannot define its own benchmark. The penultimate row names every reported value still ahead of this work, including the TRANS-D3 proposed row where it applies; on FD004 this work is second overall, not first. This work is one shared model without a subset label, whereas the compared entries are subset-specific. Official C-MAPSS tests were inspected in earlier development, so all values are exploratory rather than a pristine holdout.

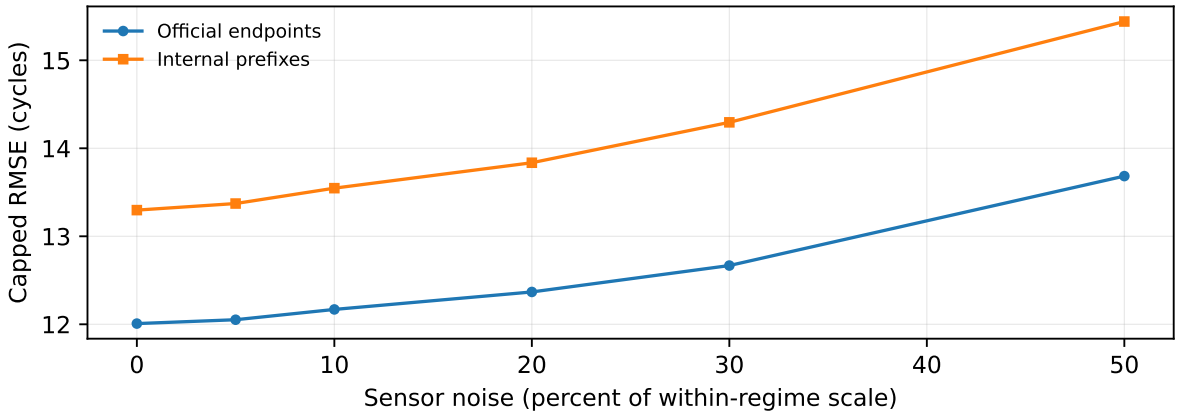


Fig. 5. Capped RMSE of the frozen artifact against broadband sensor noise. Every point is one evaluation of the same weights, scaler, corrector and interval radius; levels are paired draws, not independent realizations or training seeds.

TABLE IX
POST-FREEZE COMPONENT ABLATION OF ONE UNCHANGED ARTIFACT.

Condition	Full	−Recovery	−Corrector	Neither	Cov. (%)	Width
clean	12.01	12.00	11.99	11.98	99.3	68.2
intermittent	12.04	12.09	12.02	12.07	99.2	68.1
outage	12.46	12.81	12.43	12.81	99.1	67.8
block	12.19	12.49	12.16	12.48	99.2	68.1
noise	12.33	12.33	12.31	12.31	99.2	68.3
bias	12.09	12.14	12.07	12.13	99.2	68.3
drift	12.10	12.12	12.08	12.10	99.2	68.3
stuck	15.08	20.07	15.06	20.04	96.8	68.5
mixed	13.05	14.44	13.03	14.42	98.7	68.2

All four columns use the same weights, scaler, detector thresholds and interval radius; only the inference switches differ. Fault rows average three corruption replicates, clean has none. Nothing was selected, removed or retuned from these values, and the deployed artifact is unchanged. Coverage and mean width are for the full path under the frozen radius. Official benchmarks were inspected in earlier development, so this is a diagnostic, not a holdout.

family, but recall reaches only 0.21 for drift, so most altered entries are left for the generator rather than explicitly flagged. stuck is the easiest family (F1 0.72) and stuck the hardest to reconstruct (1.08 original-linear MAE), consistent with a held value being ambiguous rather than out of range.

Healthy-sensor protection holds: the worst intact displacement is 0.007 original-linear units and the worst false-repair fraction is 0.4%. Missing-entry families show no detector activity because they are routed by the availability mask, and their reconstruction column measures imputation of genuinely absent cells. These numbers also expose the stated limitation directly: an in-support finite fault with a low detection score is under-repaired, and coverage falls to 96.8% for stuck faults. Detection thresholds were frozen before these values were read.

H) Computational Cost

The artifact is small: 348,656 parameters, 1.33 MB of float32 weights, and 19.4 M multiply-accumulates per window. A single-window forward pass takes 5.1 ms on this CPU and the recovery path accounts for about 67% of it, which is the price of the stuck-fault gain rather than an overhead to hide. Batched throughput reaches 1285 engines per second, so the full 707-engine benchmark evaluates in 3.6 s at batch one.

TABLE X
POST-FREEZE RECOVERY AND DETECTION BENCHMARK BY FAULT FAMILY.

Family	Scope	F1	Recall	Reconst. MAE	Intact disp.	False repair (%)	Cov. (%)
clean	none	n/a	n/a	n/a	0.002	0.1	99.3
intermittent	missing	n/a	n/a	0.24	0.002	0.0	99.2
outage	missing	n/a	n/a	0.23	0.000	0.0	99.1
block	missing	n/a	n/a	0.23	0.001	0.0	99.2
noise	finite	0.37	0.23	0.15	0.003	0.1	99.2
bias	finite	0.46	0.30	0.22	0.003	0.1	99.2
drift	finite	0.35	0.21	0.16	0.003	0.1	99.2
stuck	finite	0.72	0.62	1.08	0.007	0.4	96.8
mixed	finite	0.57	0.41	0.43	0.004	0.1	98.7

Detection is measured on finite altered entries at severity 0.2, averaged over three corruption replicates; it is undefined for missing-entry families, which the availability mask handles instead, so those cells are not applicable rather than zero. Reconstruction and intact displacement are in original-linear sensor units. False repair is the fraction of healthy entries flagged. Nothing was tuned on these values and the deployed artifact is unchanged.

TABLE XI
POST-FREEZE COMPUTATIONAL COST OF THE EXPORTED ARTIFACT.

Metric	Value	Scope
Trainable parameters	348,656	whole artifact
Parameters: detector / generator / predictor / corrector	37,909 / 176,533 / 132,005 / 2,209	per module
Float32 weight size	1.33 MB	weights only
Multiply-accumulates, full path	19.4 M	one 70-cycle window
Multiply-accumulates, bypass	16.7 M	one 70-cycle window
Latency, full path	5.08 ms	single window, median
Latency, bypass	3.03 ms	single window, median
Latency, plain predictor	2.96 ms	single window, median
Throughput	1285 engines/s	batch 64, full path
Fleet pass	3.59 s	707 engines, one window each
Process peak memory	836 MB	interpreter and torch included

Wall-clock measurement of one CPU machine, not a hardware-independent claim; multiply-accumulates are a complexity proxy for supported operations only. The recovery path costs about 67% latency over bypass and buys the stuck-fault gain in Table IX. Peak memory includes the Python and PyTorch interpreter, whose baseline is reported in the saved record.

These are wall-clock numbers from one machine; latency and memory are not hardware-independent and the tables report no energy measurement.

I) Previously Inspected FD004 Engine 179

This case is a post-freeze diagnostic, excluded from the current selection objective. At cycle 240 the model predicts 7.440 against capped target 125 (raw RUL 195); its capped interval is [0.000, 22.581]. At cycle 270 it predicts 114.282. Aggregate coverage cannot establish that this particular prediction is accurate or covered. Raw settings at cycles 240 and 245 are [10.0067, 0.2516, 100] and [20.0005, 0.7, 100], respectively, and correspond to different fitted regimes. Consequently the earlier identical-setting physical-excursion explanation is unsupported. The local branch uses 30 cycles and the Transformer uses 70; a local-window-only causal explanation is also incomplete.

J) Evidence and Reproducibility Boundaries

Point metrics, counts and coverage are recomputed from saved predictions. All manuscript numbers and plotted values are tied to a single completed run, with input/output SHA-256 hashes in `paper_assets/current_provenance.json`; the noise benchmark and the cross-study transcription carry their own record in `paper_assets/current_noise_provenance.json`. Asset generation performs no model fitting or inference. The source model and earlier audit directories remain preserved; new artifacts use fresh directories. This does not restore independence of previously inspected internal or official benchmark cases. Matched reproductions of the compared methods, repeated training seeds and external fleet validation remain separate requirements.

IX. LIMITATIONS AND FURTHER VALIDATION

Validation guards protect the declared empirical contexts; acceptance does not guarantee every healthy future sensor or individual RUL prediction. In-support moves are bounded, whereas outliers may move inward across fitting support. A finite faulty in-support reading can still be under-repaired. Related-channel and temporally correlated simulation remains a hypothesis about acquisition errors, not measured aircraft fault prevalence.

A single training seed cannot establish stability. The 125-cycle target limits long-horizon raw-RUL evidence. Wide intervals can cover inaccurate point estimates. Engine exchangeability, censoring policy and fleet transfer remain assumptions; previously inspected internal and official cases remain exploratory. Neither aggregate coverage nor a plausible narrative repairs a severe individual miss. Further remedies should be designed and selected on fitting/development engines, then

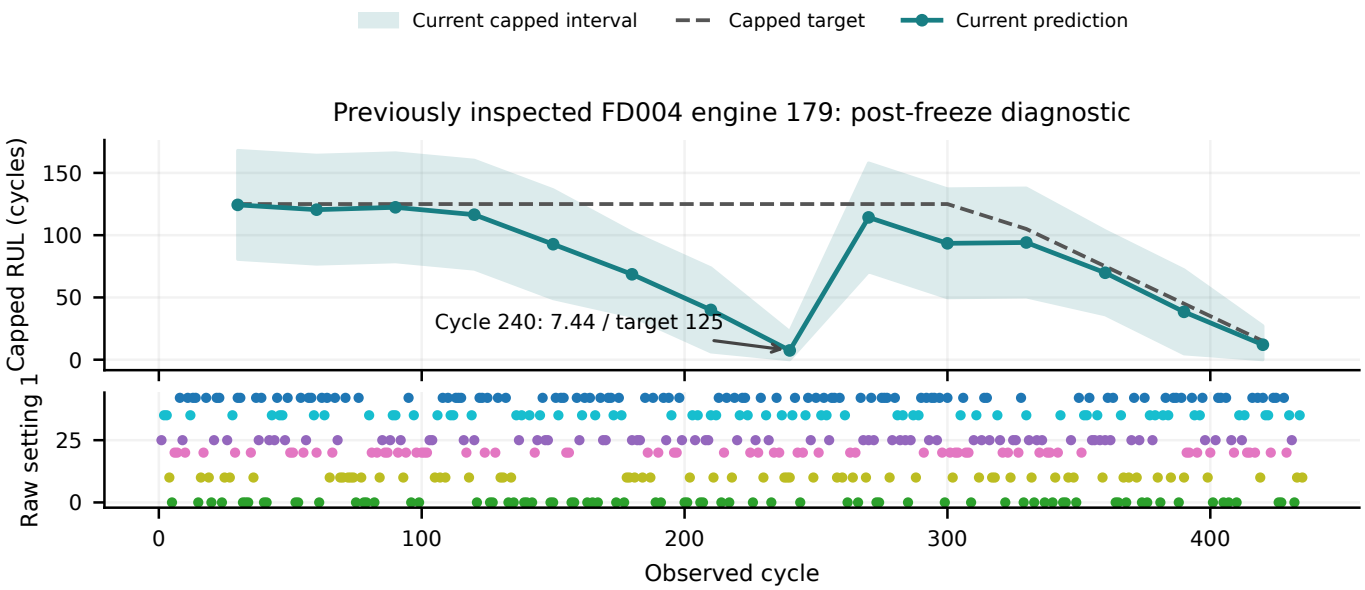


Fig. 6. Observed failure-case trajectory and operating-setting changes. Predictions and intervals are saved outputs of the frozen current artifact. Colors in the lower panel identify fitted regimes; raw settings are not ground truth about a physical fault cause.

tested on independent data. Matched paper reproductions, explanation fidelity, repeated training seeds and real-fleet validation remain incomplete. No pre-registered holdout can be reconstructed retrospectively: every C-MAPSS engine has already served in fitting, selection, calibration or evaluation, so the component ablation in Section VIII is a post-freeze diagnostic rather than independent confirmation. Optional SHAP/LIME interfaces are implemented, but this study does not report completed explanation-quality experiments. A recent hybrid Transformer–KAN–BiLSTM formulation targets the same benchmark [9]; because that evaluation re-splits merged training and test trajectories, it is cited for context only and is not used as a numeric comparator here.

X. Conclusion

The current shared GRACE-RUL pipeline combines related-channel fault simulation, independent detection, bounded recurrent recovery and local/global RUL prediction. Validation selects weights and conservative thresholds before calibration and testing. Official capped RMSE is 12.01 cycles and severity-0.2 stuck RMSE is 15.08; the full recovery matrix bounds observed intact displacement on held selection contexts. Observable scaling preserves one simultaneous calibration score per engine. Explicit failure analysis, separate coverage units and immutable provenance delimit the evidence. A post-freeze ablation attributes the persistent-fault gain to the recovery path and finds no measurable RUL benefit from the bounded corrector. These results support an auditable benchmark study; they do not establish universal sensor recovery, zero severe prediction errors, superiority over matched re-implementations of the compared methods, or real-fleet safety.

Appendix A
Evidence and Artifact Identity

The current bundle SHA-256 prefix is `577cf22aec575a40`. The source model, validation-only attempts, rejected candidates and frozen final evaluation are recorded separately. Tables, graphs and numerical prose are generated from hashed CSV/JSON evidence in `paper_assets/current_provenance.json`; no desired metric is assigned. Historical artifacts are not relabeled as current results. The source archive can retain numerical evidence and metadata without shipping raw datasets, learned weights or caches. An artifact's scope, model hash, source hashes, engine partitions and calibration contract are required to interpret its numbers.

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